

Real Repositories

<> Source Code

src/ README.rst
tests/ setup.py

Unit Tests

✓ tests/test_api.py
✓ tests/test_auth.py
✓ tests/test_client.py
✓ tests/test_utils.py

Environment Creation



SWE-agent

Attempts to install the repo
and execute tests



Developer

Writes Dockerfile based on
SWE-agent work

Task Gen. Strategies



**Procedural
Modification**



LM Generated



Combine Bugs



PR Mirroring

New Task Instances



Environment Image



Synthetic Tasks



Generated Issue



Bugged Patch +20 -12



Verified Tests ✗ 2
✓ 9

Prepare executable baseline

Mutate code, not tests

Keep new test failures